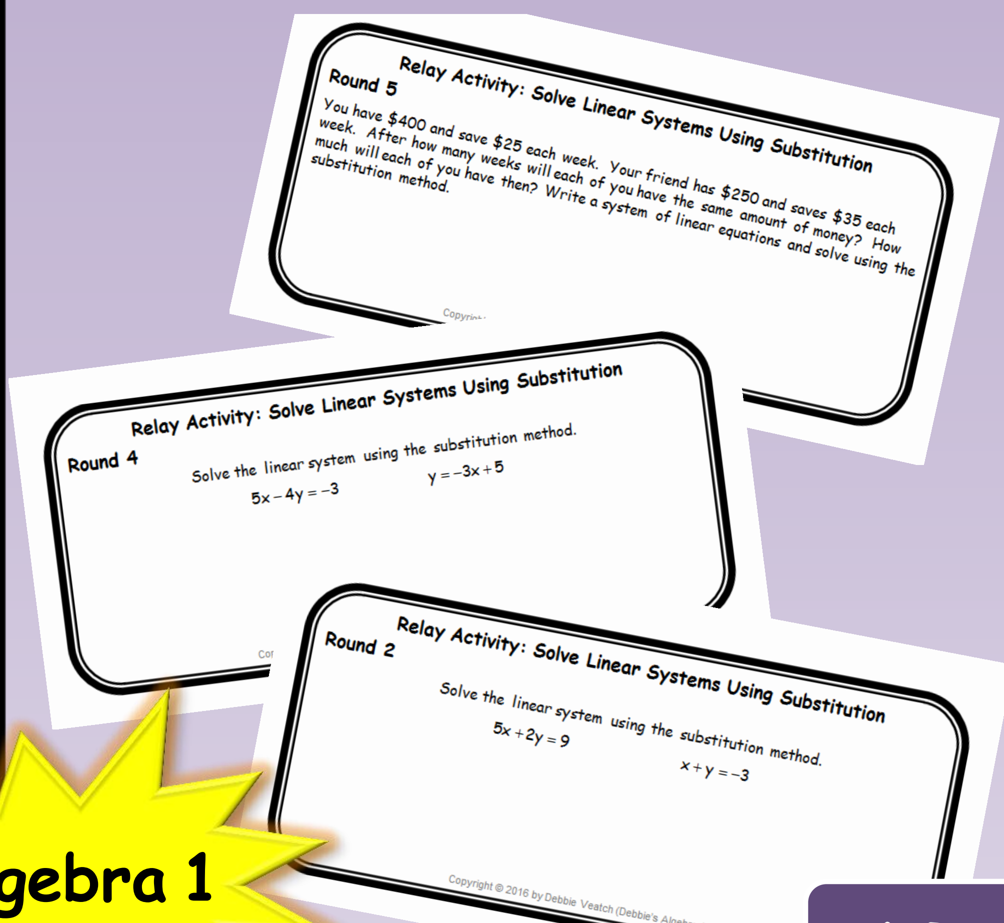
SOLVE LINEAR SYSTEMS SUBSTITUTION



Algebra 1

6 Rounds

Relay Activity

Made by Debbie Veatch

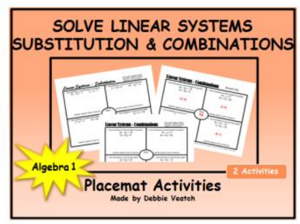
Thank You!

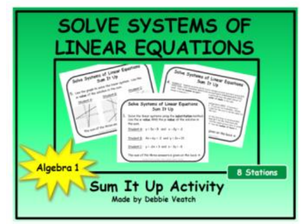
Thank you for downloading my product. I have personally used activities I have created in my own classroom and can testify to their effectiveness. I truly appreciate the support and look forward to your feedback.

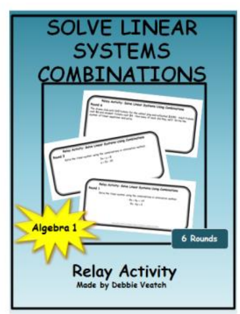
More activities are coming soon, so please check back often.

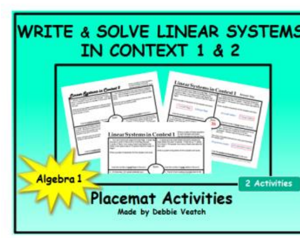
Debbie

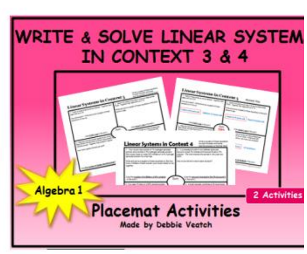
You might also like:

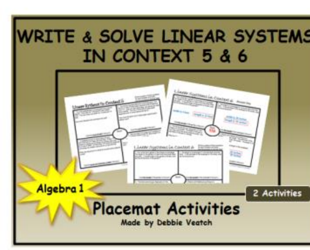












More activities coming soon. Please check back often!

Relay Activity: Solve Linear Systems Using Substitution

Objective:
To practice solving systems of linear equations using the substitution method. Activity includes a problem with "no solution" and two word problems.

CCSS: Analyze and solve pairs of simultaneous linear equations.
8.EE.C.8 Analyze and solve pairs of simultaneous linear equations.
8.EE.C.8a Understand that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because points of intersection satisfy both equations simultaneously.
8.EE.C.8b Solve systems of two linear equations in two variables algebraically, and estimate solutions by graphing the equations. Solve simple cases by inspection.
8.EE.C.8c Solve real-world and mathematical problems leading to two linear equations in two variables.
CCSS: Solve systems of equations.
HSA.REI.C.6 Solve systems of linear equations exactly and approximately (e.g., with graphs), focusing on pairs of linear equations in two variables.
CCSS: Create equations that describe numbers or relationships.
HSA.CED.A.3 Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or nonviable options in a modeling context.

Activity Directions:
Separate students into groups of four and give each student their own copy of Round 1. Students should discuss the problem as they work on it and agree on the correct answer. Once each student in the group has the correct answer, ONE person from the group takes ONE Round 1 sheet to have it checked by the teacher.

If the answer is correct, the student then picks up four copies of Round 2 for the group. If the answer is not correct, the teacher sends the student back so the group can correct Round 1.

Groups continue working until each round is completed and checked by the teacher for accuracy.

Continued →

Relay Activity: Solve Linear Systems Using Substitution

Teacher Setup:
Print enough copies (one-sided) for each student to have his/her own. Cut pages on the dotted lines to separate rounds into separate stacks (all of Round 1 is in one stack, Round 2 is in a different stack, etc.).

After handing out Round 1, keep stacks of Round 2, Round 3, etc. and only allow students to take the next round in order when their group is ready for it.

Included in the package:
- Six relay rounds
- Answer Key for Teacher

Relay Activity: Solve Linear Systems Using Substitution

Round 1

Solve the linear system using the substitution method.

$y = 6x - 11$ $\qquad$ $-2x - 3y = -7$

cut cut

Relay Activity: Solve Linear Systems Using Substitution

Round 2

Solve the linear system using the substitution method.

$5x + 2y = 9$ $\qquad$ $x + y = -3$

cut cut

Relay Activity: Solve Linear Systems Using Substitution

Round 3

Solve the linear system using the substitution method.

$-x + y = 3$ $\qquad$ $-3x + 3y = 4$

Relay Activity: Solve Linear Systems Using Substitution

Round 4

Solve the linear system using the substitution method.

$5x - 4y = -3 \qquad y = -3x + 5$

cut cut

Relay Activity: Solve Linear Systems Using Substitution

Round 5

You have \$400 and save \$25 per week. Your friend has \$250 and saves \$35 per week. After how many weeks will each of you have the same amount of money? How much money will each of you have then? Write a system of linear equations and solve using the substitution method.

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cut cut

Relay Activity: Solve Linear Systems Using Substitution

Round 6

Cole is seven years older than his cousin, Addison. Together, the sum of their ages is 15. How old is each person? Write a system of linear equations to represent this situation and solve using the substitution method.

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Answer Key for Teacher

Relay Activity: Solve Linear Systems Using Substitution

1. $y = 6x - 11$ $\quad -2x - 3y = -7$ $\quad$ **(2, 1)**

2. $5x + 2y = 9$ $\quad x + y = -3$ $\quad$ **(5, -8)**

3. $-x + y = 3$ $\quad -3x + 3y = 4$ $\quad$ **No solution**

4. $5x - 4y = -3$ $\quad y = -3x + 5$ $\quad$ **(1, 2)**

5. You have $400 and save $25 per week. Your friend has $250 and saves $35 per week. After how many weeks will each of you have the same amount of money? How much money will each of you have then? Write a system of linear equations and solve using the substitution method.

y = 25x + 400 **y = 35x + 250**

x = 15 weeks when both will have the same amount of money

y = $775 saved by each of you

6. Cole is seven years older than his cousin, Addison. Together, the sum of their ages is 15. How old is each person? Write a system of linear equations to represent this situation and solve using the substitution method.

c = a + 7 **c + a = 15**

Cole is 11 years old **Addison is 4 years old**